

INTERNSHIP PROJECT REPORT

EXCEL REPORT GENERATOR USING PYTHON

1. Introduction

The Excel Report Generator is a Python-based project designed to automate the process of creating structured and formatted Excel reports from input datasets. Preparing reports manually can be time-consuming and may result in formatting or calculation errors. This project simplifies the process by reading data, processing it, performing required calculations, and automatically generating an Excel report.

The system can be used for different types of datasets such as employee records, sales information, student details, or other tabular data. The generated report is organized into a clear Excel format, making the information easier to analyze and present.

2. Abstract

The main objective of the Excel Report Generator is to reduce the manual effort involved in preparing Excel reports. The project uses Python for data processing and Excel-related libraries to create and format the output report.

The system accepts structured input data such as CSV or Excel files. The data is loaded and processed using Python, after which relevant information is organized into tables. Calculations and summaries can be generated based on the dataset. Finally, the processed information is exported into an Excel file with appropriate headings, formatting, and report structure.

This project demonstrates how Python can be used to automate repetitive reporting tasks. It improves efficiency, reduces human errors, saves time, and provides a consistent format for generating reports.

3. Tools Used

- Python – Main programming language.
- Pandas – Used for reading, cleaning, processing, and analyzing data.
- OpenPyXL – Used to create, modify, format, and save Excel files.
- Jupyter Notebook / VS Code – Used for development and testing.
- Microsoft Excel – Used to view and analyze the generated reports.
- CSV/Excel Dataset – Used as input data.

4. Steps Involved in Building the Project

1. Dataset Collection: A relevant dataset was selected in CSV or Excel format.
2. Data Loading: The dataset was loaded into Python using Pandas.
3. Data Preprocessing: Missing values, duplicate records, and incorrect data were checked and handled where required.
4. Data Processing: Required calculations, filtering, sorting, grouping, and summary generation were performed.
5. Report Creation: The processed data was organized into a structured Excel report.
6. Excel Formatting: Headings, tables, column widths, number formats, and other formatting were applied using OpenPyXL.
7. Report Generation: The final report was saved as an .xlsx file.
8. Testing: The generated Excel file was checked to verify the data, calculations, formatting, and report structure.

5. Conclusion

The Excel Report Generator successfully automates the process of creating structured Excel reports from raw data. By using Python, Pandas, and OpenPyXL, the project reduces manual work and makes report generation faster and more consistent. The project also demonstrates practical applications of Python in data processing and automation. In the future, the system can be enhanced by adding graphical dashboards, automated email reporting, database connectivity, and support for multiple report formats.